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New evaluation of vertical particulate organic carbon fluxes of submillimeter-sized phaeodarians in the mesopelagic twilight zone of the western North Pacific Ocean

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Abstract

The western subarctic North Pacific is one of the areas where substantial amounts of CO₂ are photosynthetically taken up and effectively transferred to the ocean interior. Phaeodarians (marine siliceous Rhizaria) have high biomass in the western North Pacific, but most quantitative studies of their carbon biomass and vertical carbon flux to the sea-floor have been limited to phaeodarians larger than 500 µm. Here, we investigated the time-series flux of < 1 mm phaeodarian particles collected in a sediment trap deployed at a depth of 1000 m at Station K2 in the western North Pacific subarctic region from June 1, 2014 to July 6, 2015. We compared species-specific organic carbon content of phaeodarians in sediment trap samples measured using an ultra-sensitive elemental analyzer with that of living phaeodarians calculated using allometric carbon-to-volume relationships. This enabled us to estimate the remaining organic carbon content of all species of < 1 mm phaeodarians in sediment trap samples and quantified the < 1 mm phaeodarian organic carbon flux for the first time. The organic carbon flux of < 1 mm phaeodarians varied from 64 to 1364 µg m⁻² d⁻¹ during the sampling period, with a mean and standard deviation of 318 ± 266 µg m⁻² d⁻¹. The average contribution rate to the organic carbon flux of < 1 mm phaeodarians was higher for deeper dwelling groups, with the lower intermediate dwellers (750–1000 m) accounting for 41.6%, while the surface to subsurface dwellers (0–250 m) accounted for only 0.3%. The contribution of the organic carbon flux of < 1 mm phaeodarians to that of total (sum of the < 1 mm and > 1 mm fractions) particulates ranged from 1.1% to 12.5%, with an average contribution of 3.5% during the sampling period, which was higher than the average contribution of 2.5% reported for > 1 mm phaeodarians in our previous study. The contribution of the organic carbon flux of total phaeodarians to that of total particulates ranged from 2.4 to 15.1%, with an average contribution of 6.0%. This study highlights the importance of quantifying both < 1 mm and > 1 mm phaeodarian organic carbon fluxes for understanding the marine carbon cycle.

Keywords Rhizaria, Sediment trap, Carbon cycle, Silicon cycle, Biogeochemical cycle

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1 Introduction

Phaeodaria are unicellular zooplankton distributed throughout the world ocean (Takahashi and Anderson 2000); they are classified as a subclass in the class Thecofilosea of the phylum Cercozoa, the supergroup Rhizaria (Polet et al. 2004; Nakamura and Suzuki 2015; Adl et al. 2019). Phaeodaria have three morphological characteristics: (1) phaeodium (brown granular mass thought to be an assemblage of food and waste vacuoles), (2) central capsule (double-layered spherical protoplasmatic body), and (3) skeleton of hydrous amorphous silica ($\text{SiO}_2 \cdot n\text{H}_2\text{O}$, opal) (some exceptions do not have a skeleton) (Takahashi and Anderson 2000; Nakamura and Suzuki 2015). Radiolaria also belongs to the supergroup Rhizaria and has almost the same characteristics as Phaeodaria, but does not have a phaeodium. The skeletal morphology of Phaeodaria is diverse and varies widely in size among species, with small species ranging from tens to hundreds of micrometers and large species ranging from a few millimeters to several centimeters (Takahashi and Anderson 2000). The skeleton of phaeodaria is porous or pipe-like, and is much more fragile and easily dissolves than that of radiolarians, which also have a skeleton of hydrous amorphous silica (Takahashi and Hurd 2007). Because of the fragility of their skeletons, the fossil record of phaeodaria is scarce, but they have been present on Earth since at least the Late Triassic (Hori et al. 2009). Phaeodaria have been found at high biomass in some areas of the ocean, and can contribute to biogeochemical cycles as an important component of marine ecosystems (Steinberg et al. 2008; Biard et al. 2016; 2018; Ikenoue et al. 2019a; 2023; Laget et al. 2024).

Sinking of particles in the ocean is one of the key processes that transports carbon and other materials to the ocean interior, and is crucial for carbon sequestration. The process by which atmospheric CO_2 is taken up, fixed into organic carbon (OC) through phytoplankton photosynthesis, and sequestered by sinking particles is known as the biological carbon pump (Volk and Hoffert 1985). The major components of sinking particles are biogenic in origin, such as the remains of organisms and fecal pellets, and plankton with mineral shells or skeletons, such as diatoms, silicoflagellates, radiolaria, phaeodaria, coccolithophores, and foraminifera. To predict the rate of the biological carbon pump in the future, it is essential to quantify the amount of particulate organic carbon (POC) sequestered by this process and to characterize the composition of the biological communities that produce the sinking POC.

The western subarctic North Pacific is one of the areas where substantial amounts of CO_2 are photosynthetically taken up and effectively transferred to deep waters (Takahashi et al. 2002; Honda 2003). Although there have been

few studies on the processes that transport POC in the mesophotic twilight zone depths of 200–1000 m (Buesseler and Boyd 2009), careful sorting by microscopic examination and elemental analysis revealed that >1 mm phaeodarians in the western North Pacific account for 1.1% to 9.7% of the total POC flux at 1000 m depth (Ikenoue et al. 2019a). The carbon content of protists is proportional to their cell volume, and that of a single small protist is generally trace (Menden-Deuer and Lessard 2000). Few studies have quantified species-specific OC content in living <1 mm phaeodarians (e.g., Laget et al. 2023). Moreover, their OC content in sinking particles is often further reduced by mineralization, making accurate measurement difficult even for pooled individuals. To overcome this problem, we selected specimens of representative phaeodarian species at each layer of the water column based on their vertical distribution patterns in the plankton tows. We quantified the remaining OC content of the selected phaeodarian species in the sediment trap samples from 1000 m depth by using an ultra-sensitive elemental analyzer (Ogawa et al. 2010). For the first time, we estimated the species-specific <1 mm phaeodarian OC flux by comparing the remaining OC content of phaeodarian species from the sediment trap samples with the pre-sinking OC content of those species, which was estimated empirically from equivalent spherical diameters.

2 Methods/Experimental

2.1 Sediment trap mooring

Sinking particles were collected in a sediment trap (McLane Mark 7G-21, aperture area: 0.5 m²; Honjo and Doherty 1988) moored at Station (Stn.) K2 in the western North Pacific (47 °N, 160 °E; seafloor depth 5280 m) for 400 days from 1 June 2014 to 6 July 2015 (Table S1 and Fig. 1). The trap was deployed at a fixed depth (1000 m) and collected sinking particles into 20 bottles that rotated at consecutive intervals of 20 days. To preserve the trapped particles, each sample bottle was filled with pre-filtered deep-sea water with 5% formalin (pH neutralized) before deployment. The mooring deployment and recoveries were operated from the research vessel *Hakuho Maru* of the Japan Agency for Marine-Earth Science and Technology (JAMSTEC) during Cruises KH-14-2 and KH-15-J01. In the laboratory, sediment trap samples were first water-sieved through a 1 mm mesh nylon or plastic screen to separate larger particles such as copepods and large phaeodarians (>1 mm) from microplankton (<1 mm). The <1 mm fraction was then split into 1/100 aliquots using a rotary splitter (McLaneTMWSD-10) for enumeration and OC analysis of phaeodarians. Details of all procedures for the >1 mm fraction have been previously reported by Ikenoue et al. (2019a). The total mass

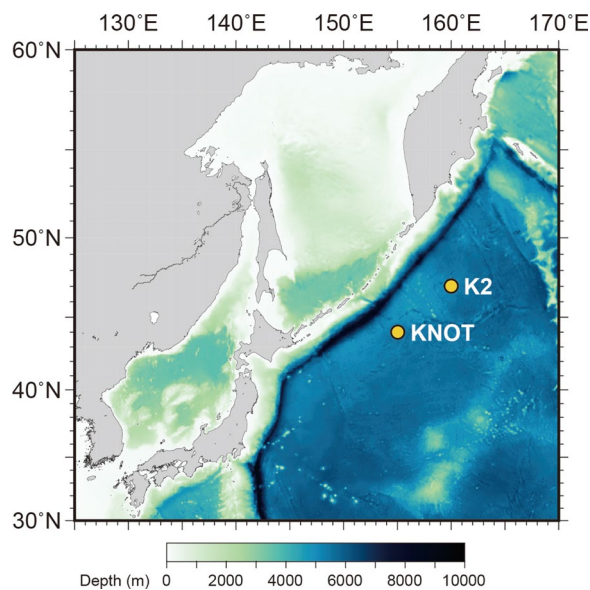


Fig. 1 Map of the study area. The yellow circles represent the location of Stns K2 and KNOT

and POC fluxes of the >1 mm and <1 mm fractions, and the abundance and OC fluxes of the >1 mm fraction phaeodarians are available online from Ikenoue et al. (2019a). Other chemical composition data for the <1 mm fraction are available online from the Database for time-series Stations (Stns.) K2 and S1 (JAMSTEC 2025).

Sea surface temperature (SST; °C) and mixed layer depth (MLD; m) around K2 (46.5 °N–47.5 °N, 159.5 °E–160.5 °E) were obtained from the Copernicus Marine Service Marine Data Store (2025). Sea surface chlorophyll-*a* concentration (Chl-*a*) (mg m^{-3}) and contemporaneous SST (°C), photosynthetically active radiation ($\text{Einstein m}^{-2} \text{d}^{-1}$), depth of euphotic zone (m), and day lengths were obtained from the Moderate Resolution Imaging Spectroradiometer onboard Satellite Aqua provided by the Ocean Color Data Portal of the Ocean Biology Processing Group at the National Aeronautics and Space Administration (NASA) Goddard Space Flight Center (NASA 2025). Eight-day mean chlorophyll-*a* concentrations were calculated for a $4^\circ \times 4^\circ$ square centered at Stn. K2. Primary productivity data ($\text{mg-C m}^{-2} \text{d}^{-1}$) estimated using the Vertically Generalized Production Model in Behrenfeld and Falkowski (1997) were obtained from the Ocean Productivity website provided by Oregon State University (2025).

2.2 Plankton tow sampling

Plankton samples in the water column (hereinafter plankton tow samples) were collected with a vertical multiple-plankton sampler (VMPS) to investigate the depth distribution of phaeodarians observed in sediment

trap samples. Plankton tow samples were collected at Stns. KNOT (OS10174) (44 °N, 155 °E) and K2, both in the western North Pacific (Table S2 and Table S3) (Fig. 1). The VMPS at Stn. KNOT (net mesh size: 63 μm , aperture area: 1 m^2) was towed on 27 July 2010 (UTC) within 12 discrete water layers (0–25, 25–50, 50–75, 75–100, 100–150, 150–250, 250–500, 500–750, 750–1000, 1000–1500, 1500–2000, and 2000–3000 m) during Cruise OS10-216 of Training ship *Oshoro-Marui*, Hokkaido University. Another VMPS at Stn. K2 (net mesh size: 62 μm , aperture area: 0.25 m^2) was towed on 8 July 2015 (UTC) within nine discrete water layers (0–30, 30–50, 50–100, 100–150, 150–200, 200–500, 500–1000, 1000–1500, and 1500–2000 m) during Cruise KH-15-J01 of the research vessel *Hakuho Maru*. The plankton tow samples at both stations were split equally into two or four aliquots with a Motoda box splitter (Motoda 1959) on board. The split samples were also water-sieved through a 1 mm-mesh stainless-steel screen, and the >1 mm and <1 mm fractions were fixed, with 5% buffered formalin at Stn. KNOT and with 99.5% ethanol at Stn. K2. The <1 mm fraction samples were stained on board with Rose Bengal to discriminate between living and dead specimens. During the plankton tow sampling at both stations, hydrographical data (temperature, salinity, and dissolved oxygen) were simultaneously obtained with a CTD (Conductivity Temperature Depth profiler) system (SBE9Plus, Sea-Bird Electronics).

2.3 Enumeration of <1 mm phaeodarian taxa

The split sediment trap and plankton tow samples were sieved through a stainless-steel screen with 45 μm mesh first and then filtered through Gelman® membrane filters with a nominal pore size of 0.45 μm to retain all particles for quantitative counts. The filtered samples were desalted with Milli-Q water and dried in an oven at 50 °C overnight, and then permanently mounted with Canada balsam® on micro-slides. Each micro-slide was examined under an Olympus BX43 transmitted light microscope at 100 $\times$ to 400 $\times$ magnification for species identification and counting. All complete phaeodarian specimens larger than 45 μm on slides were counted. Only well-preserved specimens with more than 50% of their skeletons present were counted as complete specimens (i.e., fragments were excluded from the counting).

As for sediment trap samples, the phaeodarian abundance flux (individuals $\text{m}^{-2} \text{d}^{-1}$) was calculated from count data with the following formula:

$$\text{Phaeodarian abundance flux} = N \times V/S/D \quad (1)$$

where N is the counted number of phaeodarian specimens, V the aliquot size, S the aperture area of the sediment trap (0.5 m^2), and D the sampling interval (days).

The sum of the abundance fluxes of each <1 mm phaeodarian taxon for all observed taxa is defined as the <1 mm phaeodarian abundance flux.

As for plankton tow samples, specimens that clearly stained bright red were interpreted as living cells (stained) at the depth interval where they were collected, while cells that did not stain red, or were just barely stained red, were interpreted as dead cells (empty) in the same manner as in Ikenoue et al. (2019b). Counts of phaeodarian specimens were converted to phaeodarian abundance (individuals m^{-3} of sea water).

2.4 Estimation of organic carbon and nitrogen of pre-sinking phaeodarians

To estimate the biovolumes of phaeodarian taxa, which have complex shapes, their equivalent spherical diameters were calculated by using the following formula (Llopis Monferrer et al. 2022):

$$\text{ESD} = 2\sqrt{\text{Area}/\pi} \quad (2)$$

where ESD is the equivalent spherical diameter (μm), and area is the pixel area of each imaged phaeodarian specimen on its micro-slide. The pixel area of each specimen was converted to μm^2 by using ImageJ (2025). The average ESD for each phaeodarian taxon was calculated from one to ten randomly selected specimens. From the average ESD of each phaeodarian taxon, the biovolume (μm^3) was calculated using the following formula:

$$\text{Biovolume} = \frac{4}{3}\pi(0.5\text{ESD})^3 \quad (3)$$

Species-specific OC and nitrogen (N) contents of pre-sinking (living) phaeodarians were calculated from their respective biovolumes by using the following equations established by Laget et al. (2023):

$$Q_{\text{OC}} = 10^{[0.958 \pm 0.025]} V^{[0.455 \pm 0.016]} \quad (4)$$

$$Q_{\text{N}} = 10^{[0.157 \pm 0.026]} V^{[0.477 \pm 0.017]} \quad (5)$$

where Q_{OC} and Q_{N} are the OC ($\mu\text{g individual}^{-1}$) and N ($\mu\text{g individual}^{-1}$) of each specimen, respectively, and V is the biovolume (mm^3).

The phaeodarian carbon biomass ($\mu\text{g m}^{-3}$) was calculated using the following formula:

$$\begin{aligned} \text{Phaeodarian carbon biomass} \\ = Q_{\text{OC}} \times \text{stained phaeodarian abundance} \end{aligned} \quad (6)$$

The sum of the phaeodarian carbon biomass of each <1 mm phaeodarian taxon for all observed taxa is defined as the <1 mm phaeodarian carbon biomass.

2.5 Species-specific chemical analysis of phaeodarians

Phaeodarians sink after death and some of their OC is mineralized while sinking between their dwelling depth and their collection in the sediment trap at 1000 m depth. To determine the remaining OC ratios in phaeodarian specimens in the sediment trap samples, several phaeodarian specimens representative of each layer of the water column were randomly picked out of the 1/100 split sediment trap subsample with a Pasteur pipette (Fig. S1). The representative species of phaeodaria for each layer of the water column were selected based on their vertical distribution patterns in the plankton tows. Additionally, these representative species were selected for high abundance (10 or more individuals can be obtained from sediment trap samples) to ensure adequate sample mass and for relatively simple morphology, which facilitates external cleaning. The specimens were washed several times with Milli-Q water to remove formalin and other materials attached to the outer surface before being transferred to pre-cleaned, smooth-wall tin capsules (Ludi, Switzerland) and then dried and weighed. Each sample contained 1 to 20 individuals in a capsule, depending on their body size, to maintain an optimal amount for quantification and isotopic analysis as described below. The dry weights were measured using an ultra-microbalance (UMX5, Mettler Toledo K.K., Switzerland). Four Milli-Q water samples (blanks) underwent the same procedure as above but without phaeodarian specimens. The species-specific remaining OC and N contents of phaeodarians in the sediment trap samples, along with their stable isotopic compositions were determined using a modified isotope ratio mass spectrometer (Delta plus XP; Thermo Finnigan) coupled to an elemental analyzer (FlashEA1112; Thermo Finnigan) via a continuous-flow interface (ConFloIII; Thermo Finnigan). The system was specifically modified to enhance sensitivity for ultra-small sample sizes and is hereafter called a nano-EA/IRMS (Ogawa et al. 2010). Carbon and nitrogen isotope ratios ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) are expressed as per mil (‰) deviations relative to Vienna Pee Dee Belemnite (VPDB) and atmospheric N_2 (AIR) standards, respectively. Calibration was performed using inter-laboratory-determined amino acid reference materials that cover a $\delta^{13}\text{C}$ range of -26.86 to -10.27 ‰ and a $\delta^{15}\text{N}$ range of -2.89 to 22.71 ‰: BG-T (L-tyrosine), BG-A (L-alanine), BG-P (L-proline), CERKU-01 (DL-alanine), and CERKU-02 (L-alanine) (Tayasu et al. 2011). For quantitative analysis, BG-T (L-tyrosine) was used as the reference compound.

Analytical errors, estimated from repeated measurements of BG-T, were ± 0.2 ‰ for $\delta^{13}\text{C}$ (1σ , $n=7$, 1.9 – 22 $\mu\text{g C}$) and ± 0.3 ‰ for $\delta^{15}\text{N}$ (1σ , $n=8$, 0.2 – 2.8 $\mu\text{g N}$).

2.6 Estimation of the OC flux of < 1 mm phaeodarians

The phaeodarian OC flux ($\mu\text{g m}^{-2} \text{d}^{-1}$) was calculated using the following formula:

$$\begin{aligned} &\text{Phaeodarian OC flux} \\ &= Q_{\text{OC}} \times \text{Remaining OC ratio} \times N \times V/S/D \end{aligned} \quad (7)$$

where the remaining phaeodarian OC ratio is remaining OC content / Pre-sinking OC content. The remaining OC content of phaeodarians was measured by nano EA/IRMS analysis. Pre-sinking OC content of phaeodarians was estimated from Eq. (4). The sum of the phaeodarian OC flux of each < 1 mm phaeodarian taxon for all observed taxa is defined as the < 1 mm phaeodarian OC flux.

3 Results

3.1 Hydrography and temporal variations of sinking particle fluxes

Sea surface temperature around Stn. K2 varied from about 2.4 to 14.4 °C during the sampling period, with temperatures above 10 °C observed from July to September and below 4 °C from December to May (Table S4, Fig. 2a). Mixed layer depth varied between 11 and 90 m and was deeper than 50 m from November to March due to vertical mixing associated with the winter decline in SST (Table S4, Fig. 2b). Primary productivity varied from 65 to 1425 $\text{mg-C m}^{-2} \text{d}^{-1}$, with a mean and standard deviation of $349 \pm 324 \text{ mg-C m}^{-2} \text{d}^{-1}$, and exceeded $650 \text{ mg-C m}^{-2} \text{d}^{-1}$ from late August to September (Table S5, Fig. 2c). Total POC flux at 1000 m depth varied from 2.3 to 27.3 $\text{mg m}^{-2} \text{d}^{-1}$ during the sampling period, with a mean and standard deviation of $9.1 \pm 7.1 \text{ mg m}^{-2} \text{d}^{-1}$ (Ikenoue et al. 2019a) (Table S6, Fig. 2d).

3.2 Abundance fluxes of < 1 mm phaeodarians

In total, 19 phaeodarian taxa were identified in the sediment trap samples at Stn. K2 (Tables 1 and S1, Figs. 3 and 4). The number of phaeodarians counted in each sample ranged from 35 to 228 individuals (Table S1).

The abundance flux of < 1 mm phaeodarians at 1000 m depth varied from 350 to 2280 individuals $\text{m}^{-2} \text{d}^{-1}$ during the sampling period, with a mean and standard deviation of $1185 \pm 605 \text{ individuals m}^{-2} \text{d}^{-1}$ (Table S7 and Fig. 5a). Higher than average fluxes were observed in the summer to early winter (July to December 2014) and spring (March to April 2015). The phaeodarian taxa that contributed $\geq 3\%$ to the abundance flux of < 1 mm phaeodarians during the sampling period were *Borgertella caudata* (27.3%), *Porospathis holostoma* (15.8%), *Euphysetta staurocodon* (13.6%), *Challengeron vicina* (11.6%), *Protocystis harstoni* (7.1%), *Protocystis thomsoni* (5.8%), and

Protocystis tridens (3.0%). In total, these taxa accounted for 84.2% of the abundance flux of < 1 mm phaeodarians (Table S7 and Fig. 5b).

3.3 Hydrographic profiles and vertical distribution of < 1 mm Phaeodarians

Vertical profiles of temperature, salinity, and dissolved oxygen at Stn. K2 down to a depth of 2000 m and at Stn. KNOT down to a depth of 3000 m are shown in Table S8 and Table S9 and Fig. 6. The water temperature in the 5–150 m depth interval at Stn. K2 decreased from 7.8 to 2.7 °C, and that at Stn. KNOT decreased from 15.2 to 2.7 °C. In this depth interval, the water at Stn. K2 was slightly warmer than that at Stn. KNOT, and a clear temperature minimum layer was present at a depth of 100 m at both stations. In the 150–700 m depth interval, the water temperature was slightly above the minimum and within the range of about 2.7 to 3.4 °C, but below this interval, it gradually decreased to about 1.5 °C. The vertical profile of salinity was similar at both stations, increasing rapidly from 32.5 to 32.7 at the sea surface to about 33.7 at a depth of 200 m, and then gradually increasing to about 34.7 at 3000 m at Stn. KNOT. The dissolved oxygen concentration at both stations decreased rapidly from the surface to a depth of 200 m. At Stn. K2, an oxygen minimum layer of 21–24 $\mu\text{mol kg}^{-1}$ existed from a depth of 500–1000 m, and at Stn. KNOT, an oxygen minimum layer of 41–52 $\mu\text{mol kg}^{-1}$ existed from 450–1300 m depth. Below the oxygen minimum layers, the dissolved oxygen concentration gradually increased with depth.

A total of 26 phaeodarian taxa were identified in the plankton tow samples at Stns. K2 and KNOT (Tables 1, S2, and S3 and Figs. 3 and 4). The number of phaeodarians counted in each sample ranged from 18 to 345 individuals at Stn. K2 (Table S2) and from 8 to 363 individuals at Stn. KNOT (Table S3). Equivalent spherical diameters, biovolumes, and carbon and nitrogen contents of pre-sinking (living) phaeodarian taxa estimated using Eqs. (4) and (5) are shown in Table 2. The vertical distribution of phaeodarian abundance and carbon biomass at Stns. K2 and KNOT are shown in Tables S10–S13 and Fig. 7.

At Stn. K2, the abundance of stained phaeodarians ranged from 0 to 4.3 individuals m^{-3} , the relative abundance of empty specimens was very high (73–100%), and stained specimens were rarely observed. Stained specimens were mainly observed at depth intervals of 0–200, 500–1000, and 1500–2000 m (Table S10 and Fig. 7a and b). Phaeodarian carbon biomass ranged from 0 to 0.78 $\mu\text{g m}^{-3}$ (Table S11 and upper panel of Fig. 7c). At Stn. KNOT, the abundance of stained phaeodarians ranged from 1.8 to 42.2 individuals m^{-3} , with the highest values (21.7–42.2 individuals m^{-3}) in the 75–250 m depth interval, relatively high values (8.5–19.9 individuals m^{-3}) in

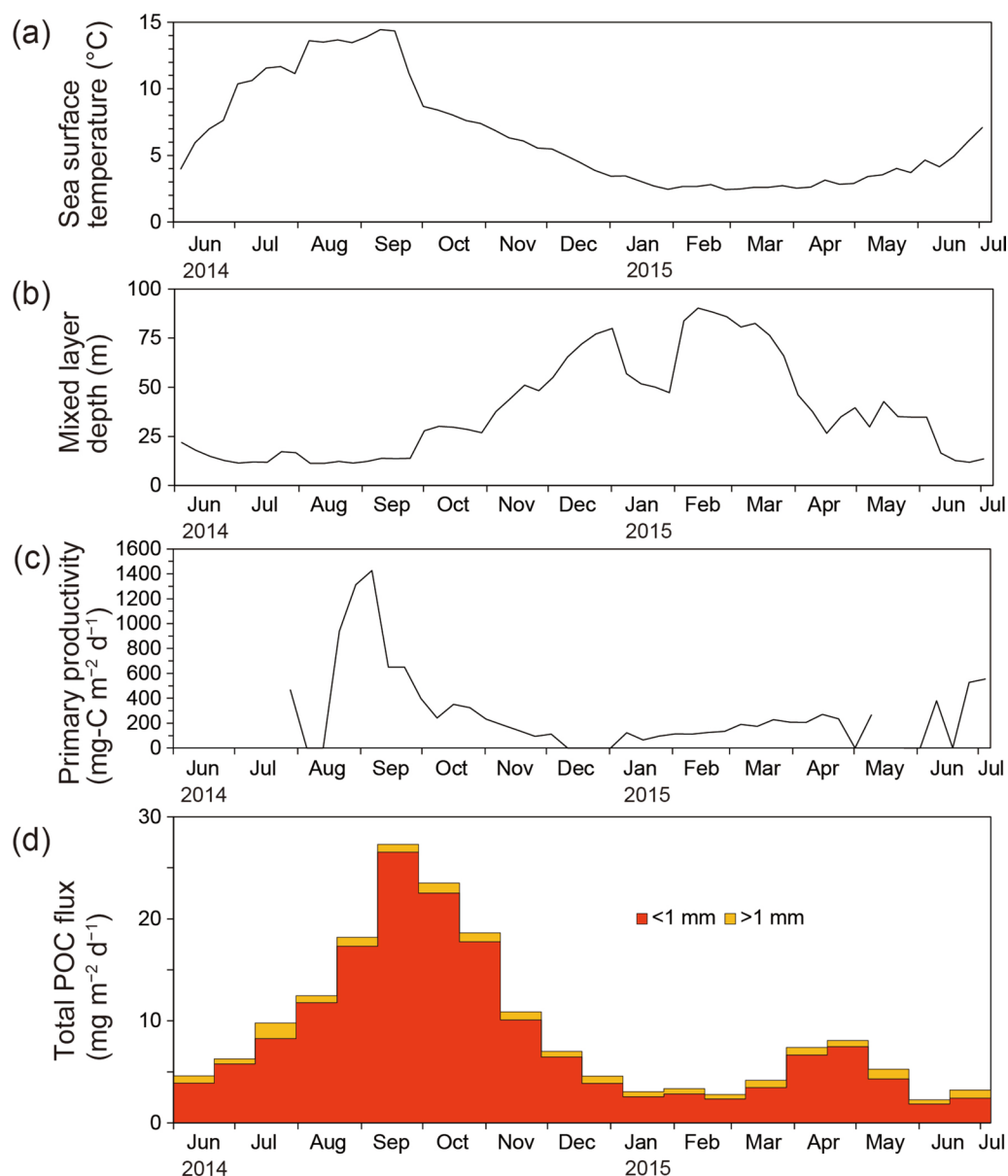


Fig. 2 Seasonal variations of hydrography and total particulate organic carbon flux at Station K2. Seasonal variations of (a) sea surface temperature, (b) mixed layer depth, (c) primary productivity, and (d) < 1 mm and > 1 mm particulate organic carbon (POC) flux

the 500–2000 m depth interval, and low values (1.8–3.8 individuals m⁻³) in the 0–75, 250–500, and 2000–3000 m depth intervals (Table S12 and Fig. 7a). Phaeodarian carbon biomass ranged from 1.08 to 21.99 $\mu\text{g m}^{-3}$, and its vertical distribution pattern was similar to that of abundance, although the highest values (15.05–21.99 $\mu\text{g m}^{-3}$) were observed in the 500–1000 m depth interval (Table S13 and Fig. 7c). The highest biomass of phaeodarians was associated with depths with relatively low temperatures (2.6–3.3 °C), high salinity (34.1–34.4), and low dissolved oxygen (41–51 $\mu\text{mol/kg}$).

The vertical distribution of stained phaeodarian abundance for each species is shown in Figs. S2–S5. Assuming that the depth interval with the highest abundance of stained specimens is the main dwelling depth interval for each species, the main dwelling depth intervals for each of the 27 taxa were classified into one of the following six categories: surface to subsurface dwellers (0–250 m), upper intermediate dwellers (250–500 m), middle intermediate dwellers (500–750 m), middle to lower intermediate dwellers (500–1000 m), lower intermediate dwellers (750–1000 m), and deep dwellers (> 1000 m) (Table 3).

Table 1 List of the 27 phaeodarian taxa encountered in sediment trap and plankton tow samples

Taxa	
Phylum	Rhizaria Cavalier-Smith (2002)
Subclass	Phaeodaria Haeckel (1879)
Order	Phaeogymnocellida Cachon and Cachon (1985) Phaeogymnocellida sp.
Order	Phaeocalpida Haeckel (1887)
Family	Castanellidae Haeckel (1879) Castanella sp. A
Family	Circoporidae Haeckel (1879) <i>Haeckeliana porcellana</i> Murray (1885)
Family	Porospathidae Borgert (1901a) <i>Porospathis holostoma</i> Cleve (1899)
Order	Phaeogromida Haeckel (1887)
Family	Challengeriidae Murray (1876), emend. Takahashi (1991) <i>Challengeron neptuni</i> Borgert (1892) <i>Challengeron tochiilinae</i> Chernyshev (2019) <i>Challengeron radians</i> Borgert (1903)* <i>Challengeron tizardi</i> Murray (1885)* <i>Challengeron ornithocephala</i> Reshetnyak (1966) <i>Challengeron vicina</i> Reshetnyak (1966) Challengeriidae sp. A* Challengeriidae sp. B* <i>Protocystis harstoni</i> Murray (1885) <i>Protocystis tridentata</i> Borgert (1903)* <i>Protocystis murrayi</i> Haeckel (1887)* <i>Protocystis thomsoni</i> (Murray 1885) <i>Protocystis tridens</i> Haeckel (1887) <i>Protocystis naresi</i> Murray (1885) <i>Entocannula infundibulum</i> Haeckel 1887*
Family	Medusettidae Haeckel 1887, emend. Takahashi (1991) <i>Euphysetta elegans</i> Borgert (1902) <i>Euphysetta staurocodon</i> Haeckel (1887) <i>Medusetta parthenopaea</i> Borgert (1901b)*
Family	Lirellidae Ehrenberg (1872) <i>Borgertella caudata</i> Wallich (1869) <i>Lirella bullata</i> Stadum and Ling (1969)** <i>Lirella melo</i> Cleve (1899)
Order	Phaeoconchida Haeckel (1887)
Family	Conchariidae Haeckel (1879) <i>Conchelum capsula</i> Borgert (1907) <i>Conchopsis compressa</i> Haeckel (1887)

The bold texts indicate the broad classification of taxa listed below

* Found only in plankton tow samples

** Found only in sediment trap samples

3.4 OC and N contents and stable isotope compositions of phaeodarians in sediment trap samples

Of the six dwelling depth intervals classified in Table 3, the following species were selected as representative for the five intervals shallower than 1000 m: *C. vicina* for the 0–250 m depth interval, *Challengeron ornithocephala* for the 250–500 m depth interval, *P. thomsoni* for the 500–750 m depth interval, *Conchelum capsula* for the 500–1000 m depth interval, and *Protocystis naresi* for the 750–1000 m depth interval. Organic carbon and N contents, and the isotopic compositions ($\delta^{13}\text{C}$ and $\delta^{15}\text{N}$) of the selected species collected from the sediment trap samples determined with nano EA-IRMS are shown in Table 4. The weights of the blank samples were close to zero, and the OC and N contents of the blank samples were negligible compared to those in the phaeodarian samples. For the five representative phaeodarian species that were analyzed, there were strong correlations ($R^2 > 0.99$) between dry weight and number of individuals, OC content and number of individuals, OC content and dry weight, N content and number of individuals, and N content and dry weight (Fig. S6). Excluding the sample “*Protocystis naresi* 1”, OC and N contents per individual of each phaeodarian species were all lower than those of the same pre-sinking phaeodarian species (Table 4). The sample “*Protocystis naresi* 1” contained more than twice the average content of OC of other specimens, so it was excluded from the calculation of the average OC content per individual of *P. naresi*. The ratios of remaining OC to that before sinking varied from 0.01 to 0.96, and were lower for species living in shallower water depth intervals (Table 5). The $\delta^{13}\text{C}$ and $\delta^{15}\text{N}$ values for each species ranged from -24.7 to -26.5 and from 4.4 to 7.2 , respectively (Fig. S6). The amount of carbon and $\delta^{13}\text{C}$ values in our samples would be slightly affected by the formalin fixation applied (e.g., Enge et al. 2018), which involves the reaction of formaldehyde and some amino acids to form cross-links in and between proteins. In addition, formalin-soluble components would be removed during the procedure. In contrast, it is empirically known that the nitrogen content and $\delta^{15}\text{N}$ value are little affected by formalin fixation (e.g., Ogawa et al. 2001).

3.5 OC fluxes of < 1 mm phaeodarians

Based on the classification of the vertical distribution (Table 3), the pre-sinking carbon content of phaeodarian taxa (Table 2) was multiplied by the remaining OC ratio of phaeodarians (Table 5) to calculate species-specific remaining OC contents of phaeodarians in the sediment trap samples. The OC flux of < 1 mm phaeodarian at 1000 m depth varied from 64 to $1364 \mu\text{g m}^{-2} \text{d}^{-1}$ during the sampling period, with a mean and standard

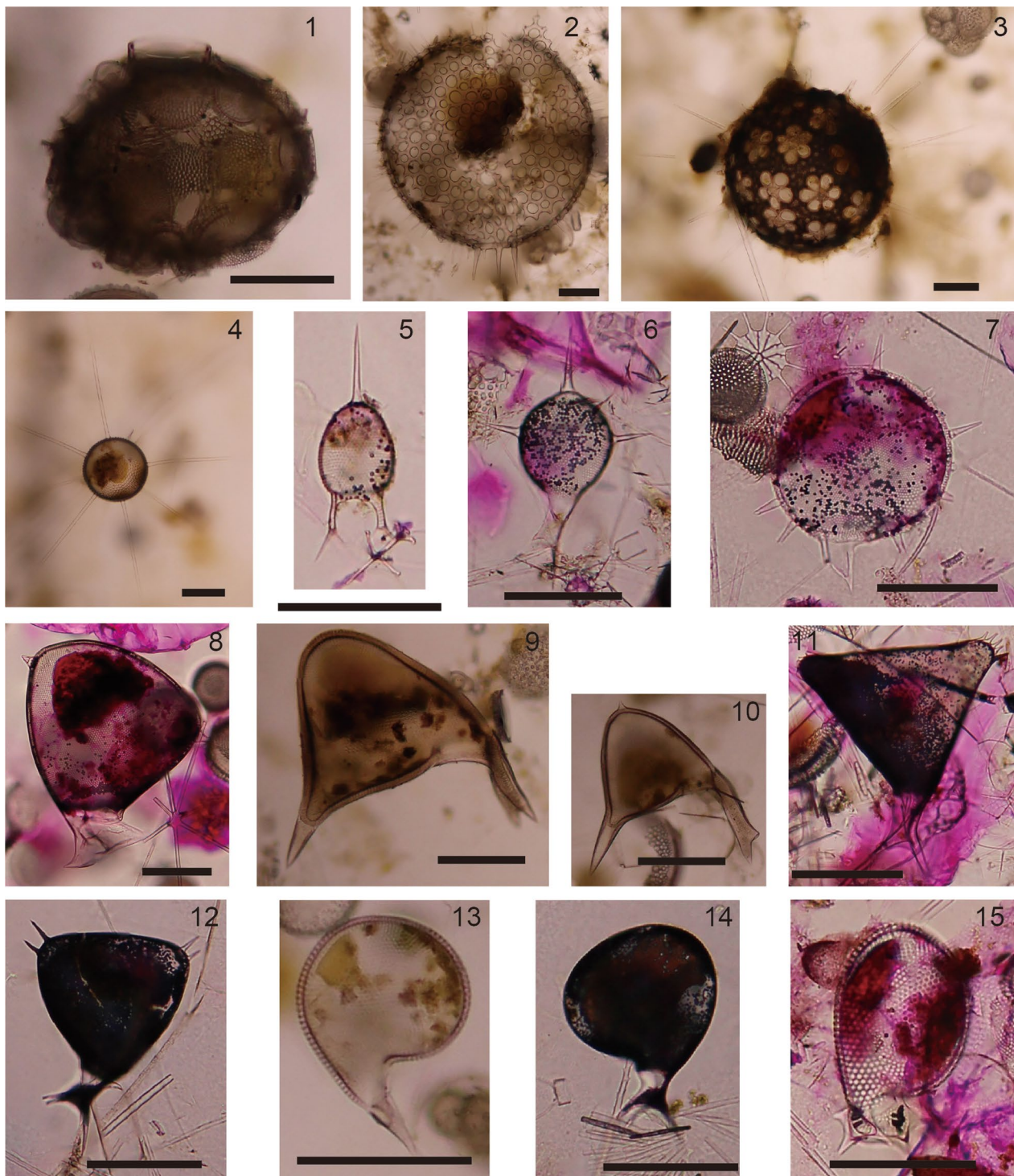


Fig. 3 Light micrographs of phaeodarian taxa identified in this study. (1) *Phaeogymnocellida* sp., sample #11; (2) *Castanella* sp. A, Stn. K2, sediment trap sample #17; (3) *Hackelia* *porcellana*, Stn. K2, sediment trap sample #7; (4) *Porospathis holostoma*, Stn. K2, sediment trap sample #3; (5) *Challengeron neptuni*, Stn. K2, plankton tow sample 150–200 m; (6) *Challengeron tochiinae*, Stn. K2, plankton tow sample 50–100 m; (7) *Challengeron radians*, Stn. KNOT, plankton tow sample 750–1000 m; (8) *Challengeron tizardi*, Stn. KNOT, plankton tow sample 2000–3000 m; (9) *Challengeron ornithocephala*, Stn. K2, sediment trap sample #11; (10) *Challengeron vicina*, Stn. K2, sediment trap sample #2; (11) *Challengeriidae* sp. A, Stn. K2, plankton tow sample 500–1000 m; (12) *Challengeriidae* sp. B, Stn. K2, plankton tow sample 1000–1500 m; (13) *Protocystis harstoni*, Stn. K2, sediment trap sample #10; (14) *Protocystis tridentata*, Stn. K2, plankton tow sample 1000–1500 m; (15) *Protocystis murrayi*, Stn. KNOT, plankton tow sample 1500–2000 m. Scale bars = 100 μ m for all figures

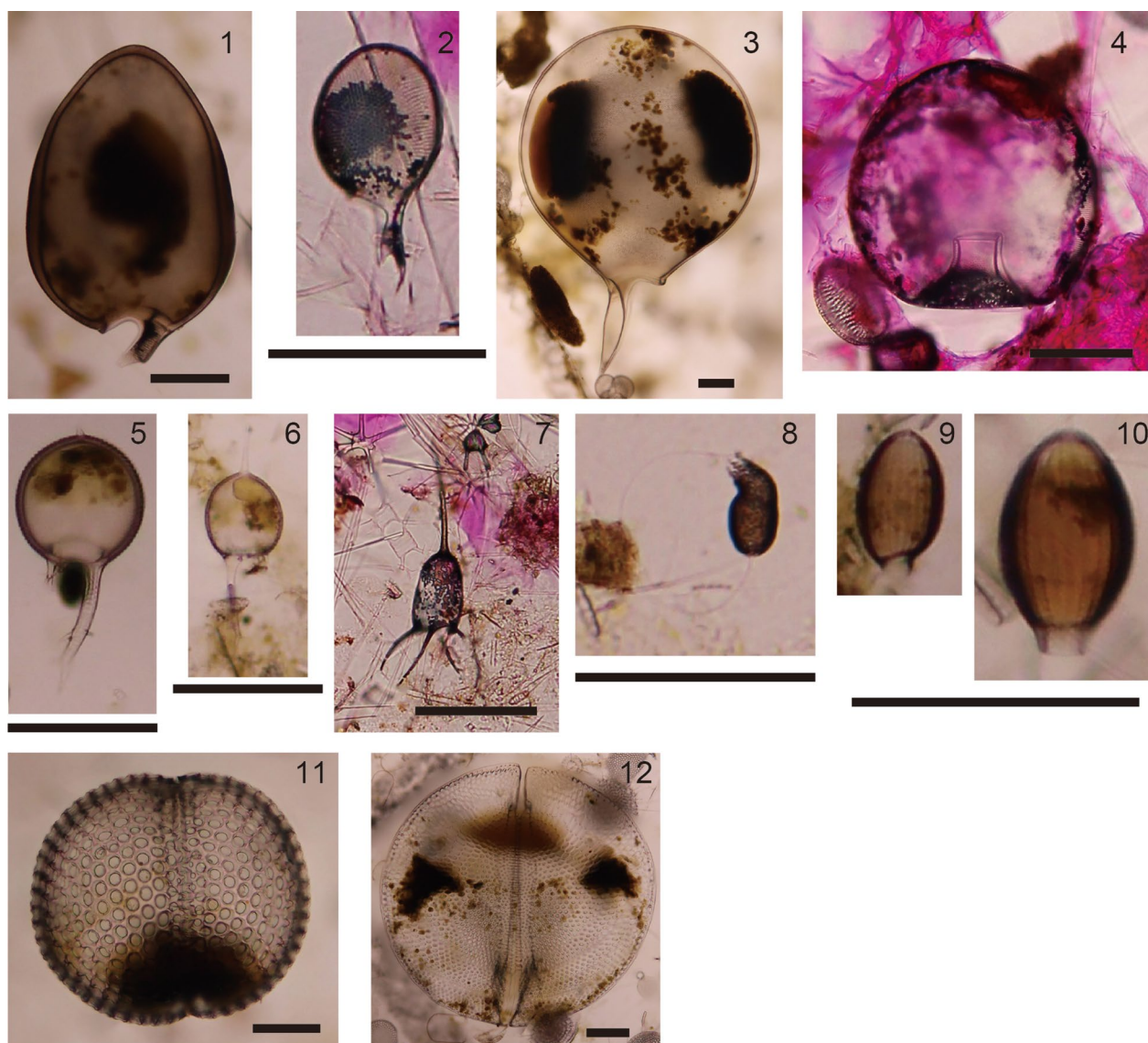


Fig. 4 Light micrographs of phaeodarian taxa identified in this study. (1) *Protocystis thomsoni*, Stn. K2, sediment trap sample #2; (2) *Protocystis tridens*, Stn. K2, plankton tow sample 50–100 m; (3) *Protocystis naresi*, Stn. K2, sediment trap sample #9; (4) *Entocannula infundibulum*, Stn. KNOT, plankton tow sample 1000–1500 m; (5) *Euphysetta elegans*, Stn. K2, sediment trap sample #10; (6) *Euphysetta staurocodon*, Stn. K2, sediment trap sample #5; (7) *Medusetta parthenopaea*, Stn. K2, plankton tow sample 500–1000 m; (8) *Borgertella caudata*, Stn. K2, sediment trap sample #1; (9) *Lirella bullata*, Stn. K2, sediment trap sample #4; (10) *Lirella melo*, Stn. K2, sediment trap sample #16; (11) *Concheliium capsula*, Stn. K2, sediment trap sample #1; and (12) *Conchopsis compressa*, Stn. K2, sediment trap sample #1. Scale bars = 100 μm for all figures

deviation of $318 \pm 266 \mu\text{g m}^{-2} \text{d}^{-1}$ (Table S14 and Fig. 8a). Fluxes higher than average were observed in summer to early winter (July to December 2014) and spring (March to April 2015). The phaeodarian taxa that contributed $\geq 3\%$ to the $< 1 \text{ mm}$ phaeodarian OC flux during the sampling period were *P. naresi* (41.6%), *P. holostoma* (20.2%), *P. thomsoni* (9.0%), *Hackeliana porcellana* (6.2%), *P. harstoni* (5.0%), *Conchopsis compressa* (4.8%), *C. capsula* (4.5%), and *B. caudata* (4.3%), in total accounting for 95.5% of the $< 1 \text{ mm}$ phaeodarian OC flux (Table S14

and Fig. 8b). The contribution of the OC flux of $< 1 \text{ mm}$ phaeodarians to the $< 1 \text{ mm}$ POC flux ranged from 1.2 to 13.5%, with an average contribution during the sampling period of 3.8% (Fig. 8c).

4 Discussion

4.1 Mineralization of phaeodarian OC in the water column

Accurately measuring the trace carbon content of $< 500 \mu\text{m}$ phaeodarians has generally been difficult. Consequently, most quantitative research on carbon

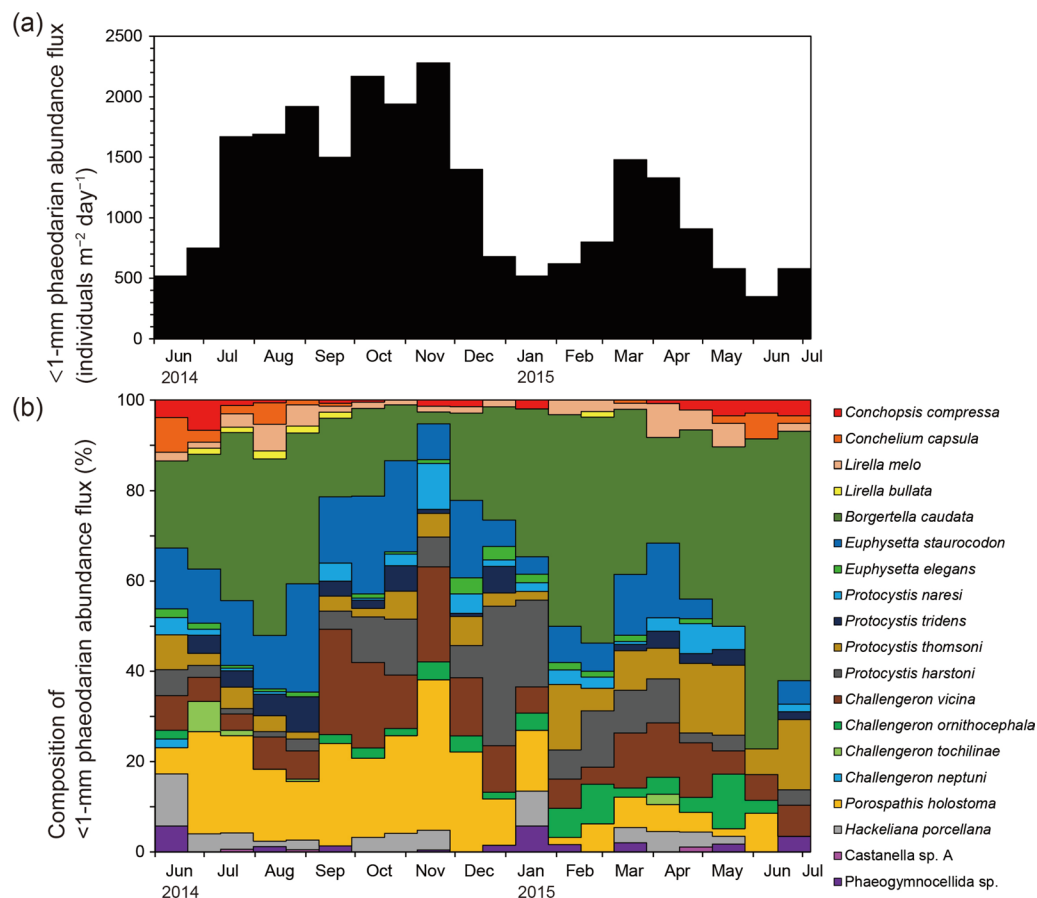


Fig. 5 Time-series of < 1 mm phaeodarian abundance flux at 1000 m depth at Strn. K2. (a) < 1 mm phaeodarian abundance flux. (b) Phaeodarian faunal compositions of < 1 mm phaeodarian abundance flux

biomass and carbon flux in phaeodarians has been limited to phaeodarians in the 500 μm to several mm size range (Ikenoue et al. 2019a; Laget et al. 2023). Recent improvements to the geometry-based formula for estimating the elemental content of Rhizaria have made it possible to accurately estimate the carbon contents of pre-sinking phaeodarians based on the individual's equivalent spherical diameter (Biard et al. 2018; Llopis Monferrer et al. 2022; Laget et al. 2023). However, since OC in sinking particles is attenuated by chemical, biological, and physical processes during gravitational sinking through the water column, geometry-based calculations alone cannot be used to directly estimate the OC content of phaeodarian individuals in sediment trap samples where some of the OC has been mineralized. To overcome this uncertainty and achieve the first estimation of species-specific OC fluxes for all species of < 1 mm phaeodarians, we have combined geometric estimation of pre-sinking OC in phaeodarians with determination of remaining OC in phaeodarians in the sediment trap samples by using an ultra-sensitive

elemental analyzer. The mineralization of organic matter within the phaeodarian skeletons in the water column was more pronounced in the shallower dwelling phaeodarian species (Table 5). In contrast, in *P. naresi*, which dwells in the 750–1000 m depth interval close to the depth where the sediment trap was deployed, only an average of 4% of its organic carbon content decrease compared to when it was alive (Table 5). OC content decreases can also be confirmed by the appearance of the phaeodarian specimens in the sediment trap: for example, the skeletons of sediment trap specimens of *C. vicina*, which dwell in the surface layer, contained almost no organic matter. The deeper the phaeodarians dwelt, the better preserved the organic matter in the sediment trap (Fig. S1). The vertical attenuation rate of POC flux in the ocean follows an empirical formula called the Martin curve, and the difference in the vertical attenuation rate between ocean regions is expressed by the Martin b parameter (exponent of the Martin curve b : $\text{POC}_{(z)} = \text{POC}_{(100)} \times (z/100)^{-b}$, where $\text{POC}_{(z)}$ and $\text{POC}_{(100)}$ represent the POC flux at water

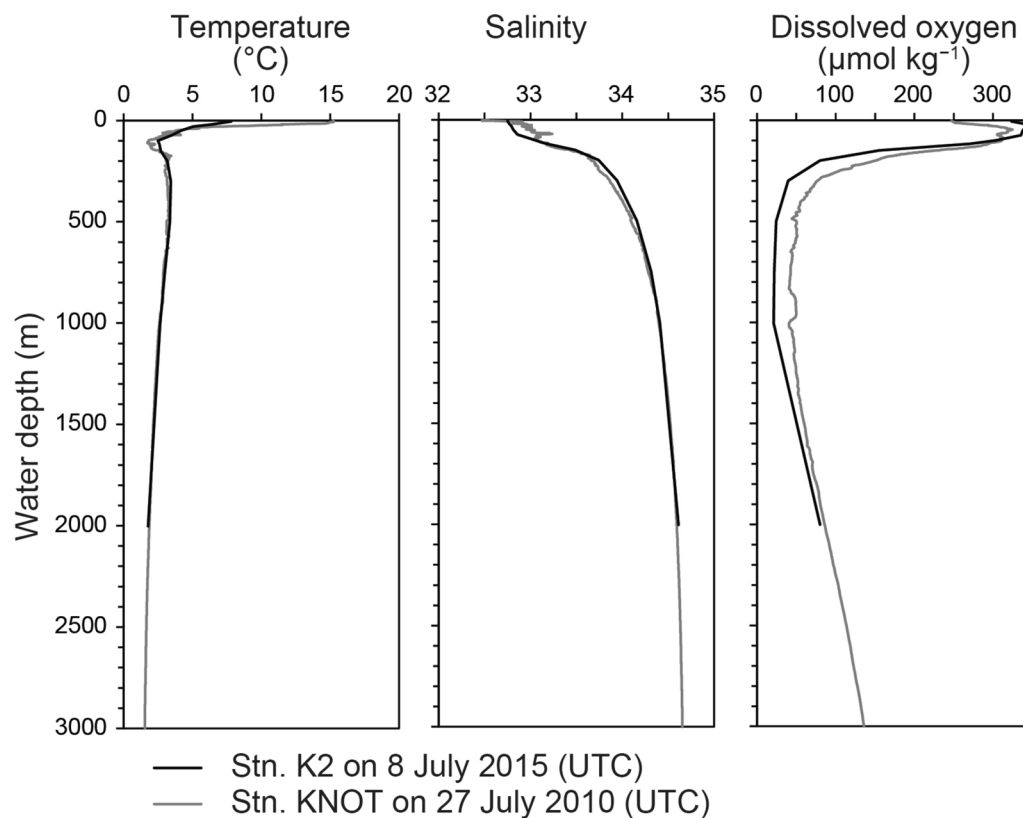


Fig. 6 Vertical profiles of temperature, salinity, and dissolved oxygen at Stns K2 and KNOT

depth z and 100 m, respectively) (Martin et al. 1987). However, the rapid mineralization of POC in shallow waters based on the Martin curve includes not only biological activity in the ocean surface layer, but also a process by which labile organic matter is mineralized first, so a direct comparison with the remaining OC rate of phaeodarians in this study could not be made. In this study, phaeodarian specimens were randomly picked out of each sediment trap sample for the entire sampling period and their OC content was measured, so the remaining OC ratio of phaeodarians reflects the average value for the entire sampling period. However, the remaining OC ratio can change depending on the seasonal changes in the primary productivity of surface phytoplankton; the predation pressure of heterotrophic organisms such as zooplankton, bacteria and protists; and the production of mucous-like substances by phytoplankton and bacteria. Therefore, for accurate estimation of phaeodarian OC flux, seasonal changes in the remaining OC ratio should be considered. However, due to the limited availability of samples for measurement, the average value of the remaining OC ratio was used in this study. Seasonal changes in the remaining OC ratio are a topic for future research.

4.2 Contribution of phaeodarian OC flux to POC flux in the Twilight Zone

The downward transport of POC to the deep ocean by the biological carbon pump lengthens the residence time of carbon in the ocean interior, and thus contributes to a reduction in atmospheric CO_2 concentrations. The attenuation of POC flux following the Martin curve is substantial up to a water depth of 500 m (Martin et al. 1987). In numerical simulations, more than half of the CO_2 transported to depths of 200–500 m is stored for less than 100 years in most ocean areas, but CO_2 transported to deeper than 1000 m is stored for hundreds of years or longer (Siegel et al. 2021). In the western North Pacific, the carbon biomass of < 1 mm phaeodarians is abundant at water depths greater than 500 m (Fig. 7). The average contribution rate for each dwelling depth group to the abundance flux of < 1 mm phaeodarians during the sampling period was highest for middle, middle to lower, and lower intermediate dwellers (500–1000 m) with 54.8%, followed by surface to subsurface dwellers (0–250 m) with 30.1% (Table 6 and Fig. S7). On the other hand, the average contribution rate to the OC flux of < 1 mm phaeodarians was higher for deeper dwelling groups, with the lower intermediate dwellers

Table 2 Organic carbon and nitrogen contents of pre-sinking phaeodarian taxa estimated empirically from equivalent spherical diameters

Taxa	n	ESD (μm)	Biovolume (mm^3)	Pre-sinking OC ($\mu\text{g}/\text{individual}$)	Pre-sinking N ($\mu\text{g}/\text{individual}$)
Phaeogymnocellida sp.	7	247 $\pm$ 32	7.85E-03	1.00 $\pm$ 0.1	0.142 $\pm$ 0.014
Castanella sp. A	9	625 $\pm$ 60	1.28E-01	3.56 $\pm$ 0.24	0.538 $\pm$ 0.037
<i>Hackeliana porcellana</i> Murray	10	407 $\pm$ 16	3.53E-02	1.98 $\pm$ 0.16	0.291 $\pm$ 0.024
<i>Porospathis holostoma</i> (Cleve)	10	159 $\pm$ 33	2.10E-03	0.55 $\pm$ 0.06	0.076 $\pm$ 0.009
<i>Challengeron neptuni</i>	10	60 $\pm$ 2	1.16E-04	0.15 $\pm$ 0.02	0.019 $\pm$ 0.003
<i>Challengeron tochilinae</i>	5	94 $\pm$ 1	4.40E-04	0.27 $\pm$ 0.04	0.036 $\pm$ 0.005
<i>Challengeron radians</i> *	1	157	2.03E-03	0.54 $\pm$ 0.06	0.075 $\pm$ 0.009
<i>Challengeron tizardi</i> *	4	275 $\pm$ 10	1.09E-02	1.16 $\pm$ 0.11	0.166 $\pm$ 0.016
<i>Challengeron ornithocephala</i>	10	218 $\pm$ 11	5.43E-03	0.85 $\pm$ 0.09	0.119 $\pm$ 0.013
<i>Challengeron vicina</i>	10	140 $\pm$ 4	1.44E-03	0.46 $\pm$ 0.06	0.063 $\pm$ 0.008
Challengeriidae sp. A*	1	141	1.46E-03	0.47 $\pm$ 0.06	0.064 $\pm$ 0.008
Challengeriidae sp. B*	3	131 $\pm$ 6	1.18E-03	0.42 $\pm$ 0.05	0.058 $\pm$ 0.007
<i>Protocystis harstoni</i>	10	102 $\pm$ 6	5.63E-04	0.30 $\pm$ 0.04	0.040 $\pm$ 0.006
<i>Protocystis tridentata</i> *	5	113 $\pm$ 3	7.50E-04	0.34 $\pm$ 0.04	0.046 $\pm$ 0.006
<i>Protocystis murrayi</i> *	2	130 $\pm$ 1	1.15E-03	0.42 $\pm$ 0.05	0.057 $\pm$ 0.007
<i>Protocystis thomsoni</i>	10	311 $\pm$ 7	1.57E-02	1.37 $\pm$ 0.12	0.198 $\pm$ 0.018
<i>Protocystis tridens</i>	7	71 $\pm$ 3	1.85E-04	0.18 $\pm$ 0.03	0.024 $\pm$ 0.004
<i>Protocystis naresi</i>	10	692 $\pm$ 19	1.74E-01	4.09 $\pm$ 0.26	0.623 $\pm$ 0.042
<i>Entocannula infundibulum</i> *	1	245	7.74E-03	0.99 $\pm$ 0.1	0.141 $\pm$ 0.014
<i>Euphysetta elegans</i>	6	97 $\pm$ 2	4.85E-04	0.28 $\pm$ 0.04	0.038 $\pm$ 0.005
<i>Euphysetta staurocodon</i>	7	58 $\pm$ 2	1.04E-04	0.14 $\pm$ 0.02	0.018 $\pm$ 0.003
<i>Medusetta parthenopaea</i> *	2	61 $\pm$ 3	1.21E-04	0.15 $\pm$ 0.02	0.019 $\pm$ 0.003
<i>Borgertella caudata</i>	10	34 $\pm$ 2	2.09E-05	0.07 $\pm$ 0.01	0.008 $\pm$ 0.002
<i>Lirella bullata</i> **	5	37 $\pm$ 2	2.71E-05	0.08 $\pm$ 0.01	0.010 $\pm$ 0.002
<i>Lirella melo</i>	10	64 $\pm$ 5	1.34E-04	0.16 $\pm$ 0.02	0.020 $\pm$ 0.003
<i>Conchelum capsula</i>	10	382 $\pm$ 12	2.91E-02	1.82 $\pm$ 0.15	0.266 $\pm$ 0.023
<i>Conchopsis compressa</i>	7	662 $\pm$ 18	1.52E-01	3.86 $\pm$ 0.25	0.585 $\pm$ 0.040

Note. Equivalent spherical diameters (ESD) and organic carbon (OC) and nitrogen (N) contents are expressed as mean $\pm$ standard deviation

* Found only in plankton tow samples

** Found only in sediment trap samples

(750–1000 m) accounting for 41.6%, whereas the surface to subsurface dwellers (0–250 m) accounted for only 0.3% (Table 6 and Fig. S8). These facts suggest that phaeodarians that dwell in water depths greater than 500 m can contribute to the sequestration of POC in water depths greater than 1000 m by delivering POC from the mesopelagic twilight zone, but that surface to subsurface dwellers, even if abundant, make almost no contribution. Phaeodarians are heterotrophic organisms that feed on particulate organic matter that sinks from the upper layers (Nimmergut and Abelmann 2002; Ikenoue et al. 2012), and they can be major producers of minipellets (González 1992). Therefore, they may play a role in transforming the POC in the intermediate layer into particles that are available for other microorganisms to consume through their feeding process, and for sinking to deeper layers.

The contribution of the OC flux of <1 mm phaeodarians to the total POC flux ranged from 1.1 to 12.5%, with an average contribution of 3.5% during the sampling period (Fig. 9b), which was higher than the average contribution of 2.5% reported for >1 mm phaeodarians (Ikenoue et al. 2019a). The total phaeodarian OC flux in this study is expressed as the sum of the <1 mm phaeodarian OC flux and the >1 mm phaeodarian OC flux. The total phaeodarian OC flux during the sampling period varied between 160 and 1640 $\mu\text{g m}^{-2} \text{d}^{-1}$, with a mean and standard deviation of 546 $\pm$ 300 $\mu\text{g m}^{-2} \text{d}^{-1}$ (Table S6 and Fig. 9a). The contribution of the total phaeodarian OC flux to the total POC flux ranged from 2.4 to 15.1%, with an average contribution of 6.0% during the sampling period (Fig. 9b). The >1 mm phaeodarians are easily attached to detritus particles on the surface of their skeleton after death and therefore contribute to the vertical

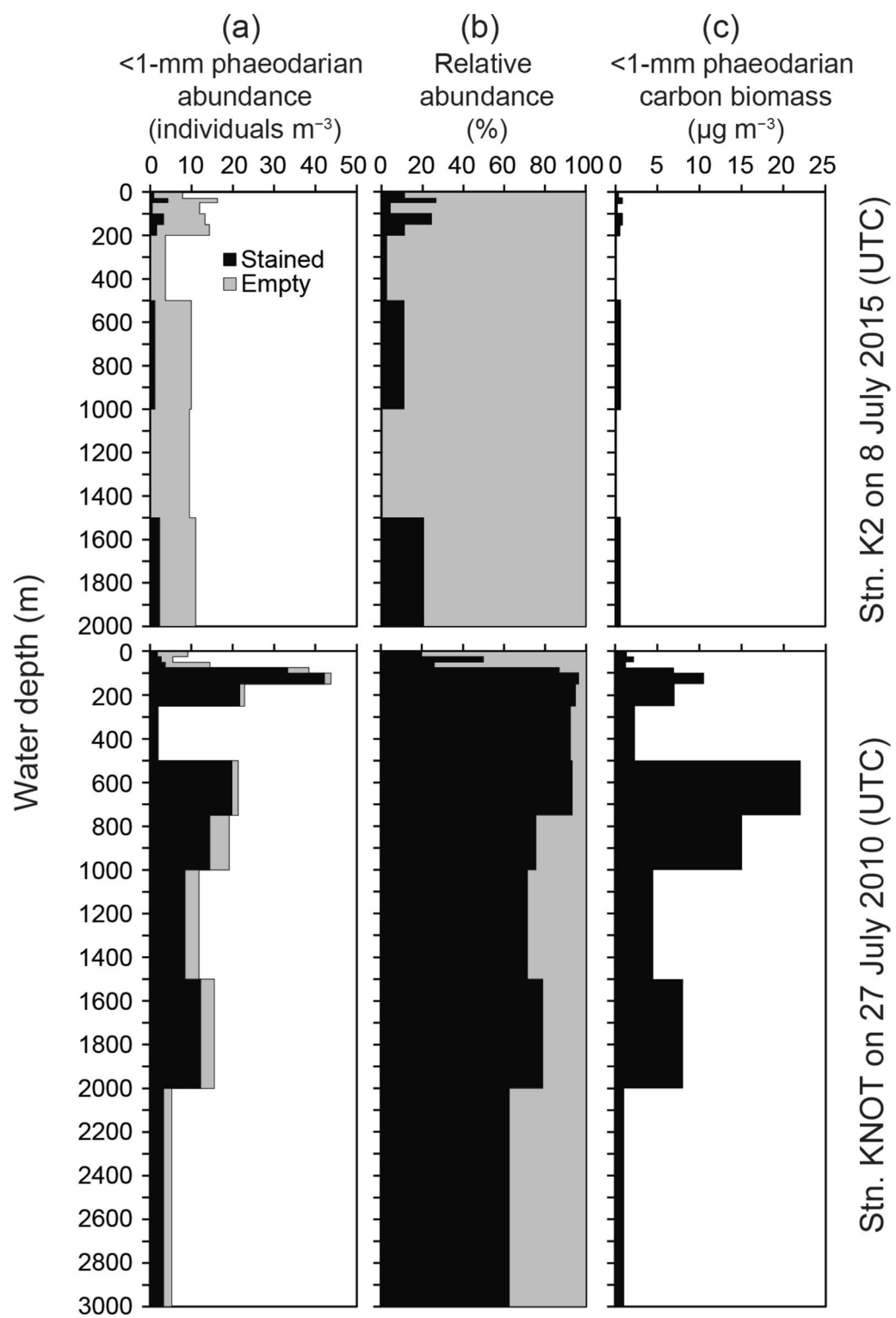


Fig. 7 Vertical distributions of < 1 mm phaeodarians at Stations K2 and KNOT. (a) Phaeodarian abundance. (b) Relative abundance of stained and empty phaeodarians. (c) Phaeodarian carbon biomass

Table 3 Phaeodarian taxa characterized by vertical distributions based on plankton tow samples

Representative depth interval of occurrence for the taxa	Taxon name
Surface to subsurface dwellers (0–250 m)	<i>Challengeron vicina</i> <i>Challengeron tochilinae</i> <i>Challengeron neptuni</i> <i>Protocystis tridens</i> <i>Euphysetta staurocodon</i> <i>Euphysetta elegans</i> <i>Lirella bullata</i> *
Upper intermediate dweller (250–500 m)	<i>Challengeron ornithocephala</i>
Middle intermediate dwellers (500–750 m)	<i>Protocystis thomsoni</i> <i>Castanella</i> sp. A <i>Hackeliana porcellana</i> <i>Conchopsis compressa</i> <i>Challengeriidae</i> sp. A
Middle to lower intermediate dwellers (500–1000 m)	<i>Conchelum capsula</i> <i>Phaeogymnocellida</i> sp. <i>Lirella melo</i> <i>Borgertella caudata</i> <i>Protocystis harstoni</i> <i>Porospathis holostoma</i>
Lower intermediate dwellers (750–1000 m)	<i>Protocystis naresi</i> <i>Challengeron radians</i>
Deep dwellers (> 1000 m)	<i>Medusetta parthenopaea</i> <i>Entocannula infundibulum</i> <i>Protocystis murrayi</i> <i>Challengeron tizardi</i> <i>Challengeriidae</i> sp. B <i>Protocystis tridentata</i>

* Since this species was not encountered in the plankton tow samples of this study, its vertical distribution was characterized based on Tanaka and Takahashi (2008)

transport of OC in other detritus particles (Ikenoue et al. 2019a). The contribution of phaeodarians including OC adhering to >1 mm phaeodarian skeletons to the total POC flux ranged from 3.1 to 17.2% during the sampling period, with an average of 8.0% (Fig. 9). The amounts of particles adhering to >1 mm phaeodarian skeletons reported in Ikenoue et al. (2019a) are the minimum amounts remaining on the surface of the skeletons after being water-sieved through a 1 mm mesh stainless-steel screen; therefore, the actual volume of particles adhering to >1 mm phaeodarian skeletons during sinking can be even greater. Since OC adhering to <1 mm phaeodarian skeletons was not studied in this study, its contribution to the total POC flux is unknown and needs to be clarified in future studies. On the other hand, <1 mm phaeodarians are so small that they not only attach detritus particles to their skeletons, but can also be packaged in fecal pellets of other large zooplankton such as copepods and sinking particle aggregates. In this study, <1 mm phaeodarians were more abundant than >1 mm phaeodarians

and accounted for a higher proportion of total POC flux. Therefore, the skeletons of <1 mm phaeodarians may contribute more to the downward transport efficiency of sinking particles as ballast than >1 mm phaeodarians. The average abundance flux of <1 mm phaeodarians at a water depth of 4800 m of Stn. K2 from March 2005 to May 2006 was 174 individuals $\text{m}^{-2} \text{d}^{-1}$ (Ikenoue et al. 2010), which is only 14.6% of the abundance flux at a water depth of 1000 m in this study. Therefore, OC and silica derived from phaeodarians in waters shallower than 1000 m depth should be mostly dissolved in seawater in the 1000–4800 m depth interval. For other sediment trap stations in the subarctic North Pacific, the average abundance flux of <1 mm phaeodarians was reported to be 1400 individuals $\text{m}^{-2} \text{d}^{-1}$ at 3200 m water depth at Stn. AB, 600 at 4810 m at Stn. SA, 680 at 3260 m at Stn. 50N, and 1150 at 2957 m at Stn. KNOT (Okazaki et al. 2005; Ikenoue et al. 2012). The phaeodarian abundance flux at approximately 3000 m of Stns. AB and KNOT is the same as or greater than that at 1000 m of Stn. K2, so more

Table 4 Remaining organic carbon (OC) and nitrogen contents, $\delta^{13}\text{C}$, and $\delta^{15}\text{N}$ of phaeodarians in the sediment trap

Sample name	Number of specimens	Dry weight (μg)	Dry weight ($\mu\text{g}/\text{individual}$)	Remaining OC ($\mu\text{g}/\text{analysis}$)	Remaining N ($\mu\text{g}/\text{analysis}$)	Remaining OC ($\mu\text{g}/\text{individual}$)	Remaining N ($\mu\text{g}/\text{individual}$)	C/N	$\delta^{13}\text{C}$ ‰ VPDB	$\delta^{15}\text{N}$ ‰ AIR
<i>Challengeron vicina</i>	22	8.2	0.4	0.11	u.d.l.*	0.005	n.d.**	n.d.**	–	–
<i>Challengeron ornithocephala</i>	17	22.7	1.3	1.6	0.28	0.09	0.016	5.6	–25.6	6.9
<i>Protocystis thomsoni</i> 1	10	28.9	2.9	4.8	0.95	0.48	0.10	5.0	–25.7	5.6
<i>P. thomsoni</i> 2	10	30.0	3.0	4.5	0.91	0.45	0.091	5.0	–25.5	5.4
<i>P. thomsoni</i> 3	11	35.0	3.2	3.6	0.80	0.33	0.073	4.5	–25.5	5.1
<i>Conchelum capsula</i> 1	1	10.2	10.2	1.5	0.33	1.5	0.33	4.6	–25.9	7.2
<i>C. capsula</i> 2	3	24.9	8.3	3.1	0.72	1.0	0.24	4.3	–24.8	7.2
<i>C. capsula</i> 3	10	91.6	9.2	11.3	2.5	1.1	0.25	4.5	–24.7	6.3
<i>Protocystis naresi</i> 1	1	24.9	24.9	8.4	1.4	8.4	1.39	6.0	–26.3	5.1
<i>P. naresi</i> 2	3	43.0	14.3	12.1	2.1	4.0	0.71	5.6	–26.2	4.4
<i>P. naresi</i> 3	11	150.8	13.7	43.1	8.2	3.9	0.74	5.3	–26.5	5.6
Blank1 (Milli-Q)	–	–	–	–0.016	0.000	–	–	–	–	–
Blank2 (Milli-Q)	–	–	–	–0.019	0.002	–	–	–	–	–
Blank3 (Milli-Q)	–	–	–	0.027	0.001	–	–	–	–	–
Blank4 (Milli-Q)	–	–	–	0.049	0.003	–	–	–	–	–

* Undetectable Limit

** Not Detected

Table 5 Summary of remaining and pre-sinking organic carbon concentrations, and their ratios for selected phaeodarian species

Species name	Representative depth range for the species (m)	Remaining OC* (μg/individual)	Pre-sinking OC*** (μg/individual)	Remaining OC ratio	δ ¹³ C ‰ VPDB****	δ ¹⁵ N ‰ AIR****
<i>Challengeron vicina</i>	Surface to subsurface (0–250)	0.005	0.46	0.01	–	–
<i>Challengeron ornithocephala</i>	Upper intermediate (250–500)	0.09	0.85	0.11	–25.6	6.9
<i>Protocystis thomsoni</i>	Middle intermediate (500–750)	0.4	1.37	0.30	–25.5 ± 0.1	5.4 ± 0.2
<i>Conchelim capsula</i>	Middle and lower intermediate (500–1000)	1.1	1.82	0.62	–25.1 ± 0.7	6.9 ± 0.5
<i>Protocystis naresi</i>	Lower intermediate (750–1000)	3.9**	4.09	0.96	–26.3 ± 0.2	5.0 ± 0.6

* Average value obtained by dividing the total value of remaining organic carbon (OC) (μg/analysis) in Table 4 by the number of specimens

** Average excluding values for *Protocystis naresi* Sample 1 have been excluded

*** Values of pre-sinking OC in Table 2

**** Mean and standard deviation of δ¹³C and δ¹⁵N in Table 4

phaeodarian-derived OC may be transported to deeper layers at Stns AB and KNOT. On the other hand, it has been suggested that large phaeodarians (> 500 μm) attenuate POC flux by 3.8–9.2% in the mesopelagic twilight zone, where they dwell (Laget et al. 2024). Therefore, in order to estimate the carbon sequestration into the interior of the ocean due to the sinking of phaeodarians, it will also be necessary in the future to accurately quantify the biomass and seasonal changes of living phaeodarians.

4.3 Seasonal changes in phaeodarian abundance and their limiting factors

The primary productivity at Stn. K2 peaked in August and September, and the total POC flux at 1000 m water depth peaked about 20 days later (Fig. 2). The increase in the phaeodarian abundance flux was generally synchronized with the increase in the total POC flux, suggesting that they feed on the particulate organic matter that sinks from the upper layer in response to the increase in primary production there (Fig. 5). However, middle and lower intermediate dwellers (500–1000 m) increased rapidly in July, which was earlier than the peak total POC flux (Fig. S7); so, it is possible that other factors, such as changes in predation pressure and microbial biomass, also increased phaeodarians there. The phaeodarian species observed in the sediment trap samples in this study are almost the same as those reported by Okazaki et al. (2005) in the sediment trap samples from Stns. 50N (50 °N, 165 °E) and KNOT at a water depth of 3000 m (Table 1), and the species that were not observed in this study were those that dwell in waters deeper than 1000 m. The period when the abundance fluxes of < 1 mm phaeodarians at Stns. 50N and KNOT reaches its peak at 3000 m depth is from fall to winter, and is 1–2 months later than that at 1000 m depth at Stn. K2 (Fig. 5). Considering that the sinking velocity at Stns. 50N and KNOT is 115 m day^{−1} (Honda et al. 2002), the timing of the flux

peaks at both stations at 1000 m depth is 17 days earlier than those at 3000 m depth. Therefore the timing of the flux peaks at Stns. K2 and 50N is almost the same at 1000 m depth, and that at Stn. KNOT is about one month later. Therefore, the species composition of the three stations located in the western North Pacific subarctic and the patterns of seasonal variation in flux are generally similar.

The < 1 mm phaeodarians mainly dwelled in the subsurface layer (100–200 m) and the mesopelagic layer (500–2000 m) at Stns. K2 and KNOT in July (Fig. 7). The subsurface layer is characterized by water temperatures of < 3.0 °C, salinities of > 33, and dissolved oxygen concentrations of > 100 μmol kg^{−1}, whereas the mesopelagic layer is characterized by water temperatures of < 3.0, salinities of > 34, and dissolved oxygen concentrations of < 80 μmol kg^{−1} (Fig. 6). Therefore, we suggest that both the epipelagic and mesopelagic species prefer low temperatures and high salinity, but epipelagic and mesopelagic species are separated by the concentration of dissolved oxygen. Although it is unclear whether differences in dissolved oxygen concentrations directly affect the vertical separation of epipelagic and mesopelagic species, their separation may be due to changes in the characteristics of sinking food and predation pressure from other organisms in response to changes in dissolved oxygen concentrations.

δ¹³C and δ¹⁵N are affected by food sources and trophic levels, respectively (Wada 1987; Wada et al. 1987). Phaeodarians feed on particulate organic matter sinking from the upper layers (Nimmergut and Abelman 2002), which is composed of miscellaneous biogenic particles such as the carcasses of organisms, fecal pellets, and bacteria. The relationship between δ¹³C and δ¹⁵N in phaeodarian species did not change with depth, but different ratios were observed for each species (Table 5 and Fig. S6), suggesting that phaeodarians could selectively

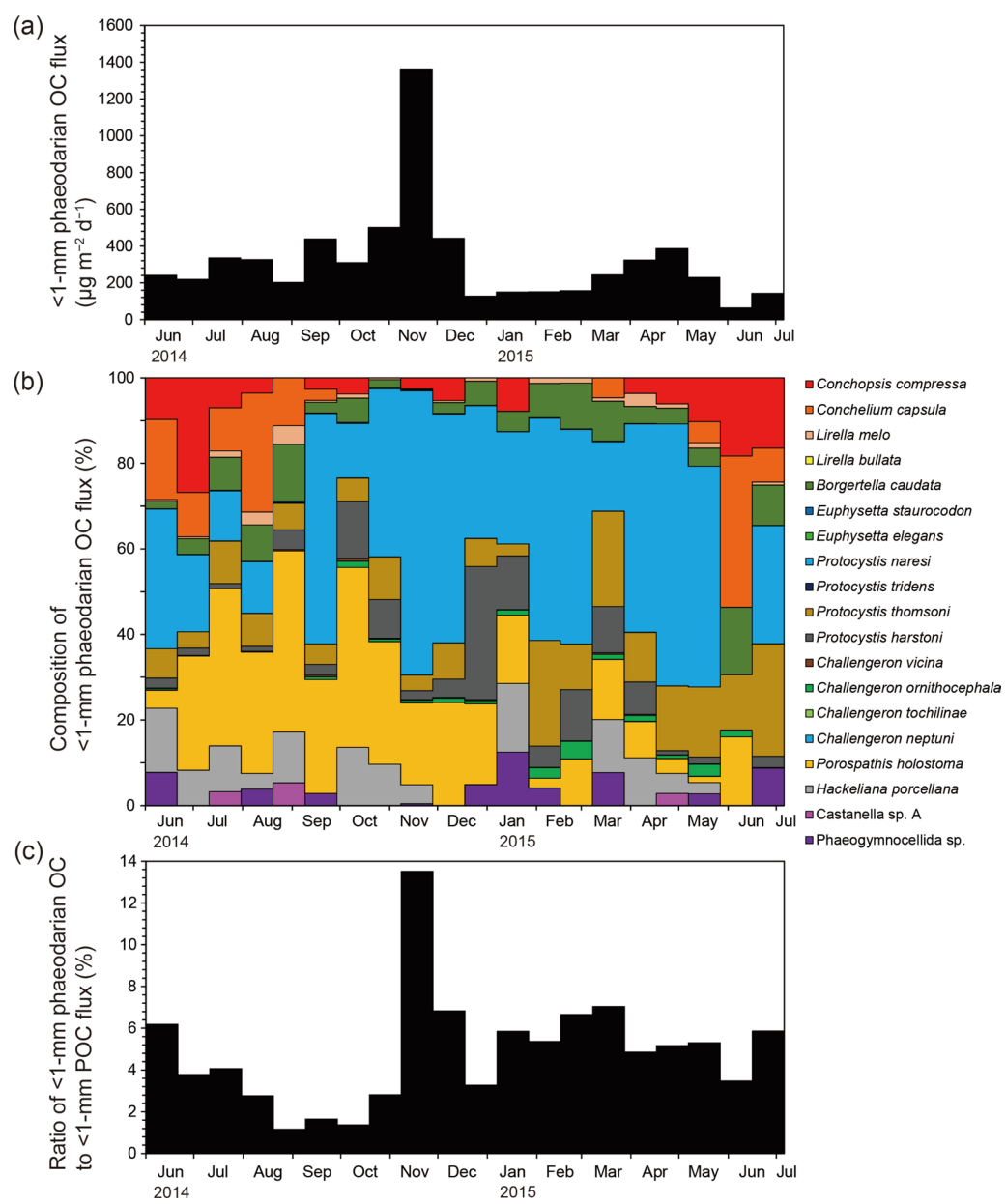


Fig. 8 Time-series of organic carbon (OC) flux of < 1 mm phaeodarians at 1000 m depth at Stn. K2. **(a)** OC flux of < 1 mm phaeodarians. **(b)** Phaeodarian faunal composition of < 1 mm phaeodarian OC flux. **(c)** < 1 mm phaeodarian OC as a percentage of < 1 mm particulate organic carbon (POC) flux

Table 6 Average contribution of phaeodarian groups by dwelling depth to phaeodarian fluxes at 1000 m depth during the sampling period

Group (m)	Contribution to < 1 mm phaeodarian abundance flux (%)	Contribution to < 1 mm phaeodarian OC flux (%)
Surface to subsurface dwellers (0–250)	30.1	0.3
Upper intermediate dweller (250–500)	2.5	0.9
Middle intermediate dwellers (500–750)	9.8	20.4
Middle to lower intermediate dwellers (500–1000)	54.8	36.8
Lower intermediate dwellers (750–1000)	2.8	41.6

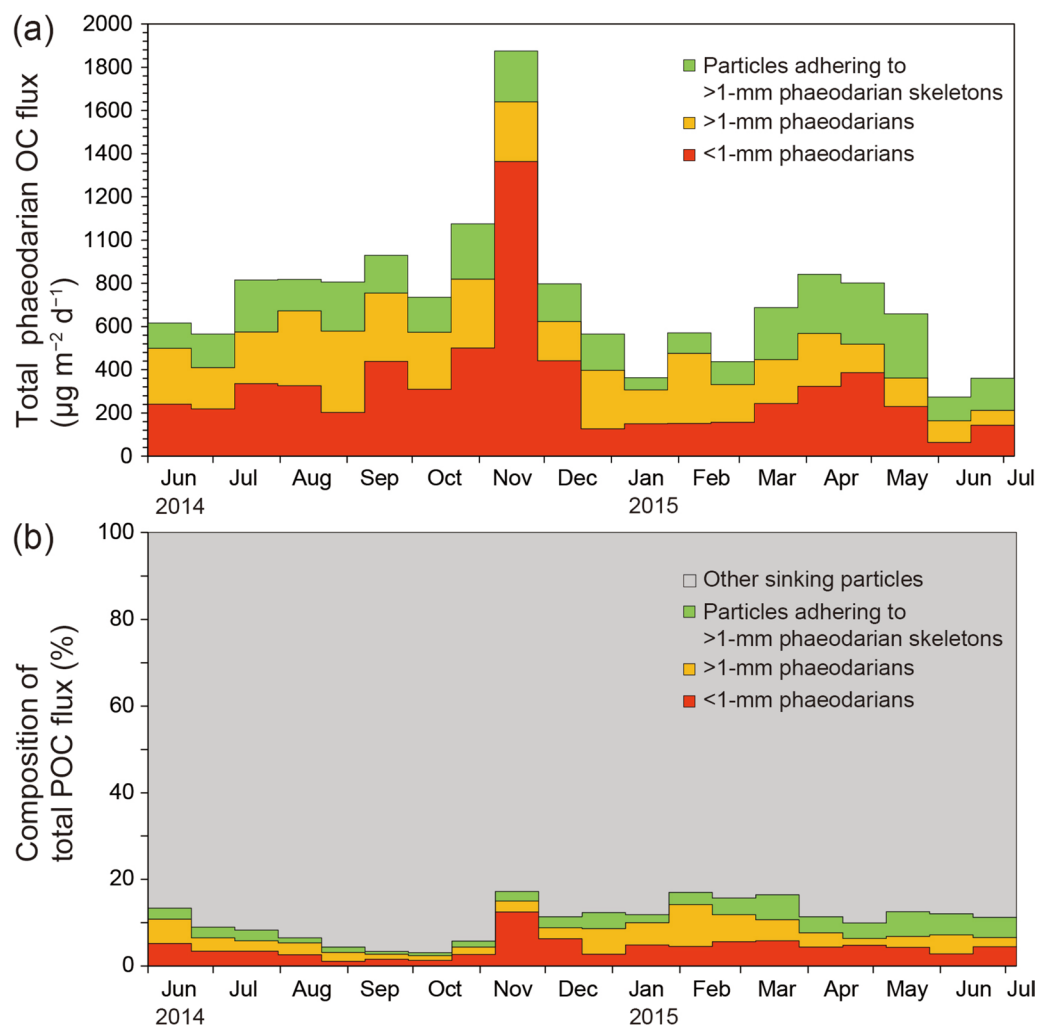


Fig. 9 Time-series of total phaeodarian organic carbon (OC) fluxes at Stn. K2. (a) Total phaeodarian OC flux. (b) Mass composition of total particulate organic carbon (POC) flux

feed on sinking particles according to species or that the ratio of grazing to detritus-feeding could differ among species.

The abundance of <1 mm phaeodarians at Stn. K2 was markedly lower than that at Stn. KNOT, with empty specimens accounting for 73–100% of the total at Stn. K2 (Fig. 7). The only marked difference in hydrography between the two stations was that the dissolved oxygen concentration in the dissolved oxygen minimum layer in the 500–1000 m depth interval at Stn. K2 was about half that at Stn. KNOT. However, the <1 mm phaeodarian abundance at Stn. K2 was markedly lower than that at Stn. KNOT even in the subsurface layer with high dissolved oxygen concentrations (Fig. 7), suggesting that that the difference in dissolved oxygen concentration was not directly related to the difference in phaeodarian abundance between the two stations. In addition, the

abundance of >1 mm phaeodarians in the same samples collected at Stn. K2 was highest in the 200–500 m depth interval, where the abundance of <1 mm phaeodarians was minimal (Ikenoue et al. 2019a). This vertical distribution could suggest competition for food and dissolved silica among different sizes of phaeodaria, or differences in their roles in the food web structure. The phaeodarian abundance flux at 3000 m water depth at Stn. KNOT was higher than that at 1000 m water depth at Stn. K2 (Okazaki et al. 2005), so we assume that the phaeodarian abundance in the water column at Stn. KNOT was also higher than that at Stn. K2. The period when plankton tow samples were collected at Stn. K2 in early July coincided with the period when the phaeodarian abundance flux was at its lowest (May to July) (Fig. 5). Therefore, the abundance of <1 mm phaeodarians at Stn. K2 in the water column was probably lower than that at Stn.

KNOT because the phaeodarian abundance in the water column was usually lower than at Stn. KNOT, and the samples were collected at a time when the phaeodarian production at Stn. K2 was low. Therefore, the markedly lower abundance of <1 mm phaeodarians in the water column at Stn K2 than that at Stn KNOT in this study was mainly due to the usually lower abundance of phaeodarians at Stn K2 than that at Stn KNOT, as well as the sampling period when the abundance of phaeodaria was low at Stn. K2.

5 Conclusions

We quantified the OC flux of <1 mm phaeodarians in the western North Pacific Ocean for the first time by combining the measurement of species-specific OC content by using an ultra-sensitive elemental analyzer with geometric estimation of OC content. The contribution of the OC flux of <1 mm phaeodarians to the vertical transport of POC was slightly higher than that of the >1 mm phaeodarians reported in our previous study, and the contribution of total phaeodarians to the vertical transport of POC in the western North Pacific was found to be up to 15.1% (average 6.0%). This study highlights the importance of quantifying both <1 mm and >1 mm phaeodarian OC fluxes for understanding the marine carbon cycle in the western North Pacific. However, such studies have not been done in other ocean basins. Therefore, it is necessary to apply the methodology of this study to sediment trap samples in other ocean basins and to quantify phaeodarian OC fluxes on a global scale. Phaeodarians, which have high biomass in the mesopelagic twilight zone, are unique organisms that can contribute to POC transport to depths of greater than 1000 m by delivering POC to depths greater than 500 m. However, they are also organisms that consume the OC of sinking particles in the mesopelagic layer. Therefore, to estimate the net carbon sequestration by phaeodarians in the interior of the ocean, it is necessary to accurately quantify the biomass of living phaeodarians and their seasonal changes in the western Pacific Ocean, which is currently an area lacking in observations. Furthermore, the trophic level and feeding habits of each phaeodarian species should be estimated by analyzing amino acid nitrogen isotope ratios, and the role of phaeodarians in the marine ecosystem as well as the global carbon cycle should be clarified.

Abbreviations

JAMSTEC	Japan Agency for Marine-Earth Science and Technology
OC	Organic carbon
POC	Particulate organic carbon
Stn.	Station
Stns.	Stations
SST	Sea surface temperature
MLD	Mixed layer depth
Chl-a	Chlorophyll-a concentration
NASA	National Aeronautics and Space Administration

VMPS	Vertical multiple-plankton sampler
CTD	Conductivity Temperature Depth profiler
ESD	Equivalent spherical diameter
N	Nitrogen
EA/IRMS	Isotope ratio mass spectrometer coupled to an elemental analyzer via a continuous-flow interface
VPDB	Vienna Pee Dee Belemnite
AIR	Atmospheric N ₂

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40645-025-00753-z>.

Additional file 1
Additional file 2
Additional file 3
Additional file 4
Additional file 5
Additional file 6
Additional file 7
Additional file 8
Additional file 9
Additional file 10
Additional file 11
Additional file 12
Additional file 13
Additional file 14
Additional file 15

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Author contributions

TI conducted treatment and analysis of phaeodarian specimens on sediment trap and plankton tow samples, drew figures, tables, interpreted the data, and wrote the manuscript. TI, KK, MCH, and TF proposed the topic, conceived, and designed the study. TI, KK, and TF collected the samples. TI, NOO, and HN carried out the experimental study. TI and NOO analyzed the data. The manuscript was written through contributions of all authors, and all authors have approved the final version of the manuscript.

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Data availability

The datasets supporting the conclusions of this article are included within the article and its additional files.

Declarations

Competing interests

The authors declare that they have no competing interest.

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